

Car Price Prediction

Regularization Analysis

A Comparative Study of Linear, Ridge & Lasso Regression

Prepared: June 30, 2026

EXECUTIVE SUMMARY

This report evaluates three regression approaches for predicting car prices: a baseline Ordinary Least Squares (OLS) model, Ridge Regression (L2 regularization), and Lasso Regression (L1 regularization). After comprehensive preprocessing including log transformation, outlier capping, feature engineering, and standardization, Ridge Regression achieved the best performance (R-squared = 0.8966). Lasso offers a compelling alternative by eliminating 50% of features while retaining 99% of Ridge performance, making it ideal for interpretability-focused applications.

1. Introduction

Accurately predicting car prices is a fundamental problem in the automotive industry, with applications ranging from dealership pricing strategies to consumer decision-making and insurance valuation. The challenge lies in the complex interplay of numerous vehicle attributes, including technical specifications, equipment features, age, mileage, and brand positioning.

This analysis applies three linear regression approaches to a car price dataset containing over 5,000 listings. The primary objectives are: (1) Build a predictive model that explains the variance in car prices, (2) Evaluate whether regularization techniques (Ridge and Lasso) improve generalization compared to a standard OLS baseline, and (3) Identify the most influential factors driving car prices for actionable business insights.

Dataset Overview

The dataset includes technical attributes (horsepower, engine size, weight, mileage), categorical classifications (make, model, fuel type, transmission, category), and equipment feature groups (comfort, entertainment, extras, safety). After preprocessing, the feature matrix contained 54 numeric predictors. The target variable is the listed car price.

2. Data Preprocessing

2.1 Missing Data & Cleaning

Columns with over 40% missing values were removed. Numerical gaps were filled with the median, and categorical gaps with the mode. This approach preserves data integrity while minimizing bias.

2.2 Feature Engineering

- Log transformation applied to price to normalize its right-skewed distribution (Figure 2).
- IQR-based capping on km, hp_kW, age, and Weight_kg to limit outlier influence (Figure 3).
- Equipment list columns (Comfort_Convenience, Entertainment_Media, Extras, Safety_Security) converted to integer count features.
- One-hot encoding of all categorical variables with drop_first=True to avoid multicollinearity.
- StandardScaler normalization applied after an 80/20 train-test split.

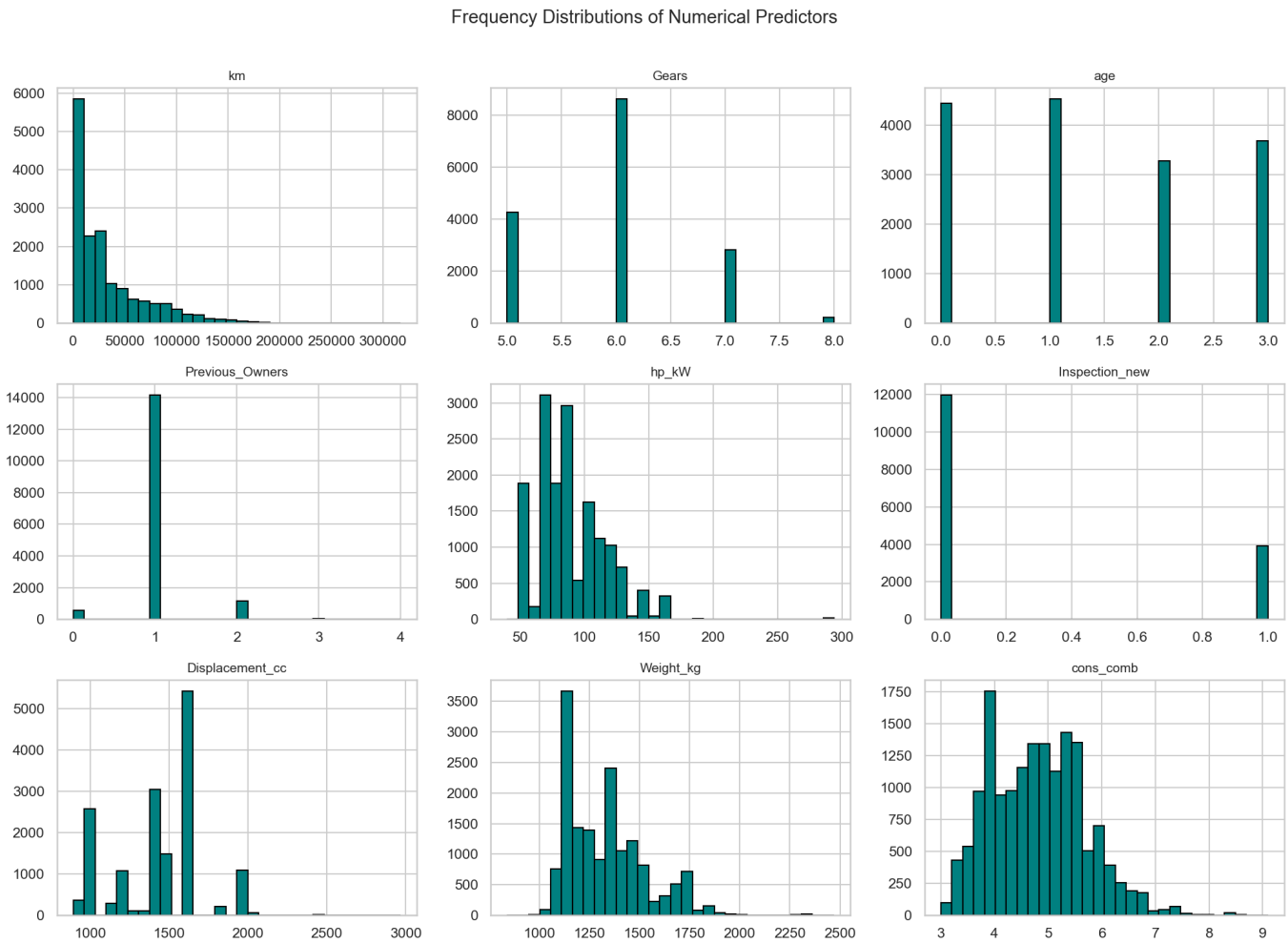


Figure 1: Frequency distributions of numerical predictors showing the spread of key features before preprocessing.

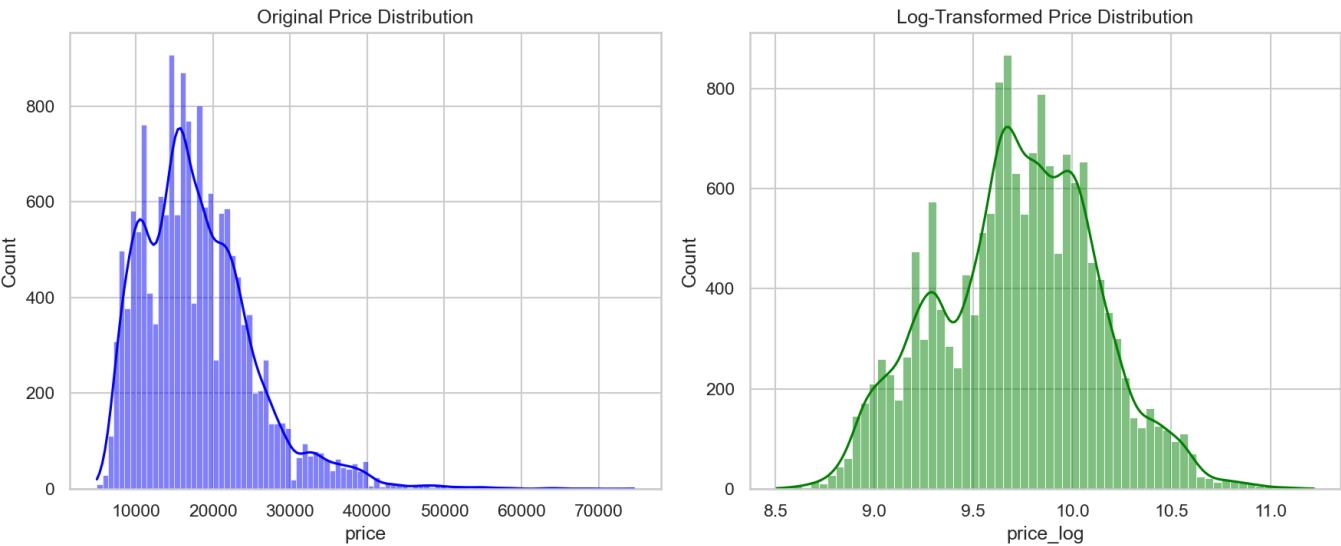


Figure 2: Log transformation of the target variable (price) normalizes a right-skewed distribution into a near-normal one.

3. Exploratory Data Analysis

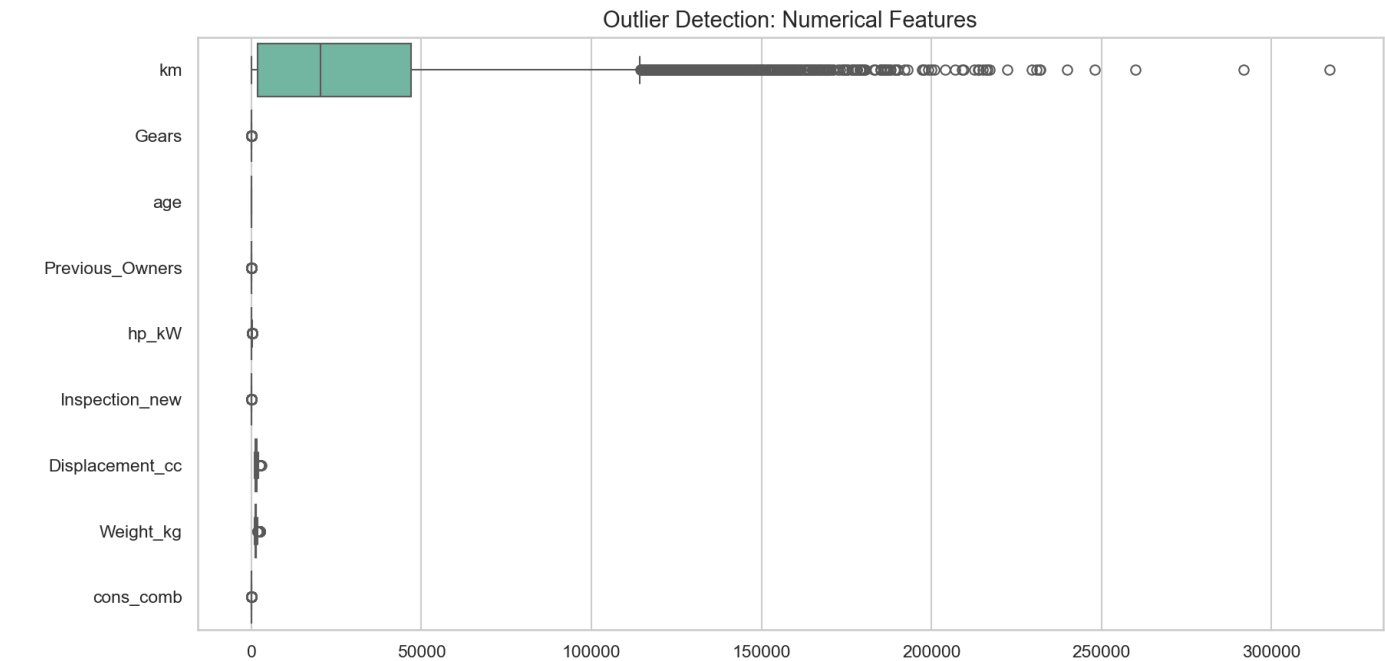


Figure 3: Boxplot analysis revealing outliers in mileage (km), horsepower (hp_kW), age, and weight.

The boxplot reveals significant outliers in mileage (km), horsepower (hp_kW), age, and weight. These extreme values can disproportionately influence regression coefficients. The IQR capping approach constrains these values to 1.5x the interquartile range, reducing their leverage while preserving all observations in the dataset.

4. Baseline Linear Regression

The Ordinary Least Squares (OLS) model serves as our performance baseline. All 54 features were included without any regularization. The model achieves the following metrics:

Model	R-squared	MAE	RMSE
Baseline OLS	0.8848	0.2606	0.3401

An R-squared of 0.8848 indicates that 88.48% of the variance in log-transformed prices is explained by the model. However, the model assigns non-zero coefficients to all 54 features, suggesting possible overfitting to noise from less informative predictors.

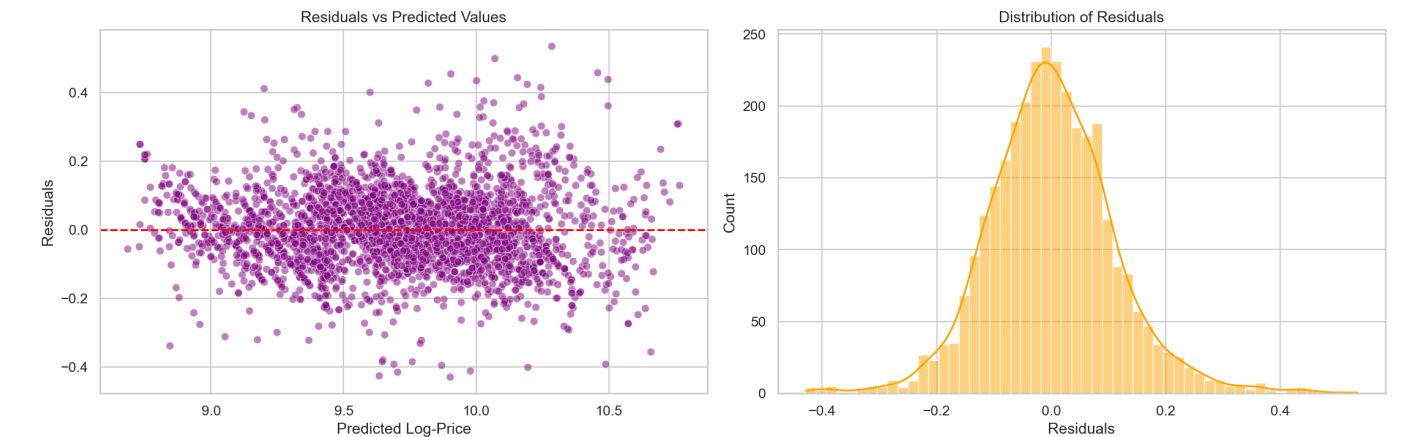


Figure 4: Residual analysis for the baseline OLS model - randomness (left) and normality (right) checks.

Residual diagnostics confirm the linearity assumption is satisfied (residuals scatter randomly around zero). The residual distribution is approximately normal, though some heavier tails are present, indicating occasional larger prediction errors.

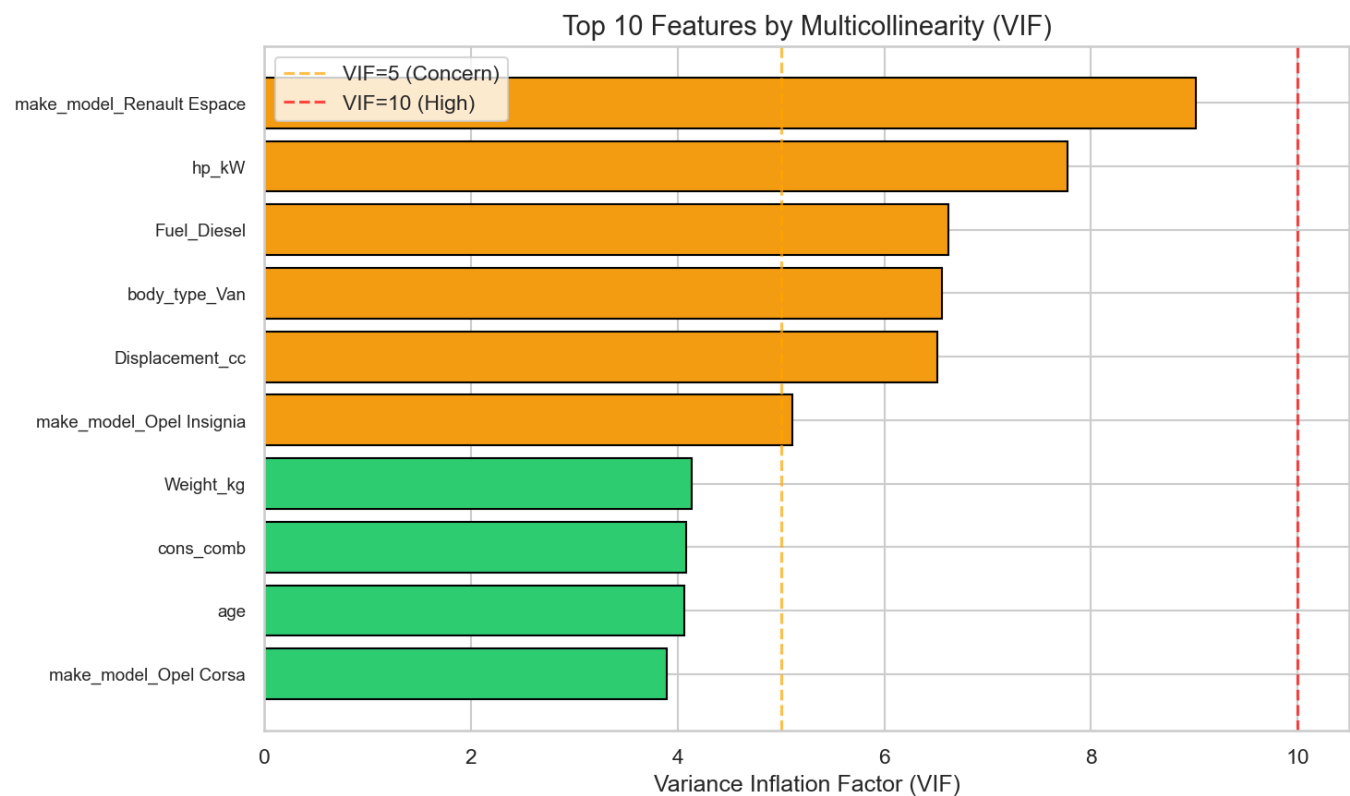


Figure 5: Multicollinearity assessment via VIF - features exceeding the threshold (red dotted line at VIF=10) indicate high collinearity.

Variance Inflation Factor (VIF) analysis reveals moderate multicollinearity among several features. Features with VIF > 5 (orange dotted line) warrant attention, while those exceeding VIF > 10 (red dotted line) indicate high collinearity that regularization can help mitigate.

5. Ridge Regression (L2 Regularization)

Ridge regression adds a penalty proportional to the squared magnitude of coefficients, shrinking all coefficients toward zero without eliminating any. This reduces model variance and improves generalization to unseen data.

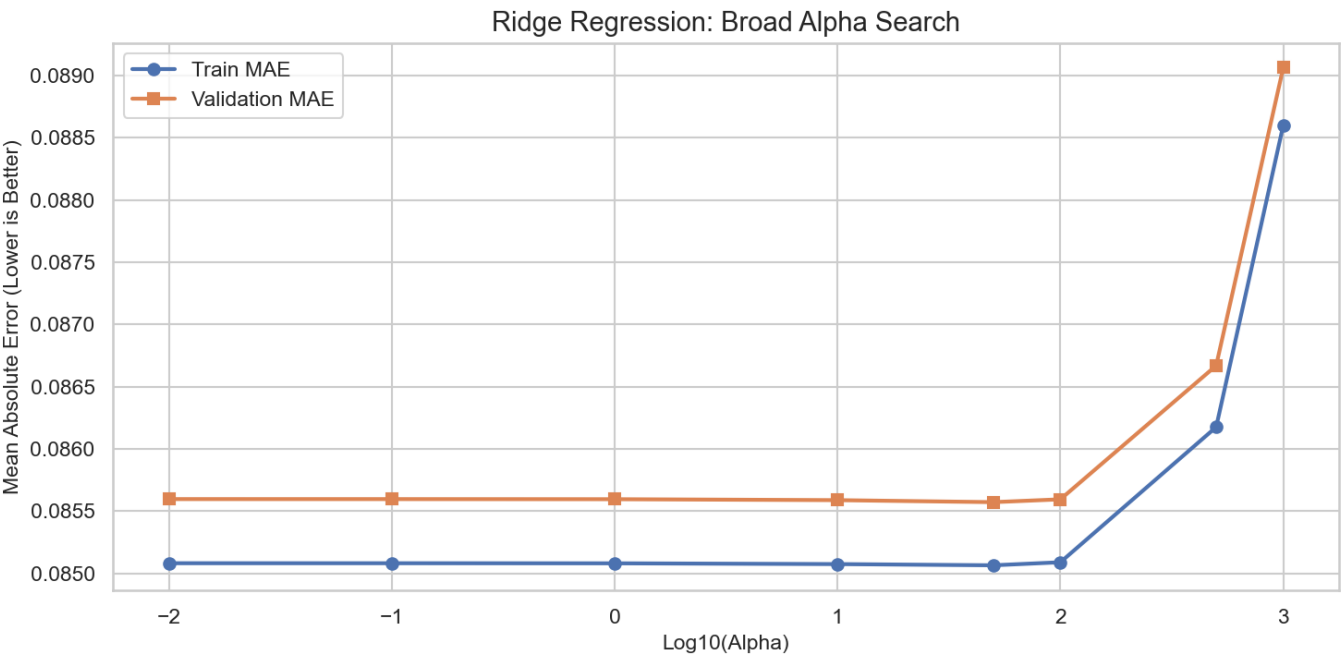


Figure 6: Ridge regression alpha tuning - validation MAE curve identifies the optimal regularization strength.

A broad alpha search across [0.01, 0.1, 1, 10, 50, 100, 500, 1000] identified the optimal region, followed by a fine-grained search yielding the best alpha of 10. The validation MAE curve shows that moderate regularization (alpha around 10) minimizes validation error.

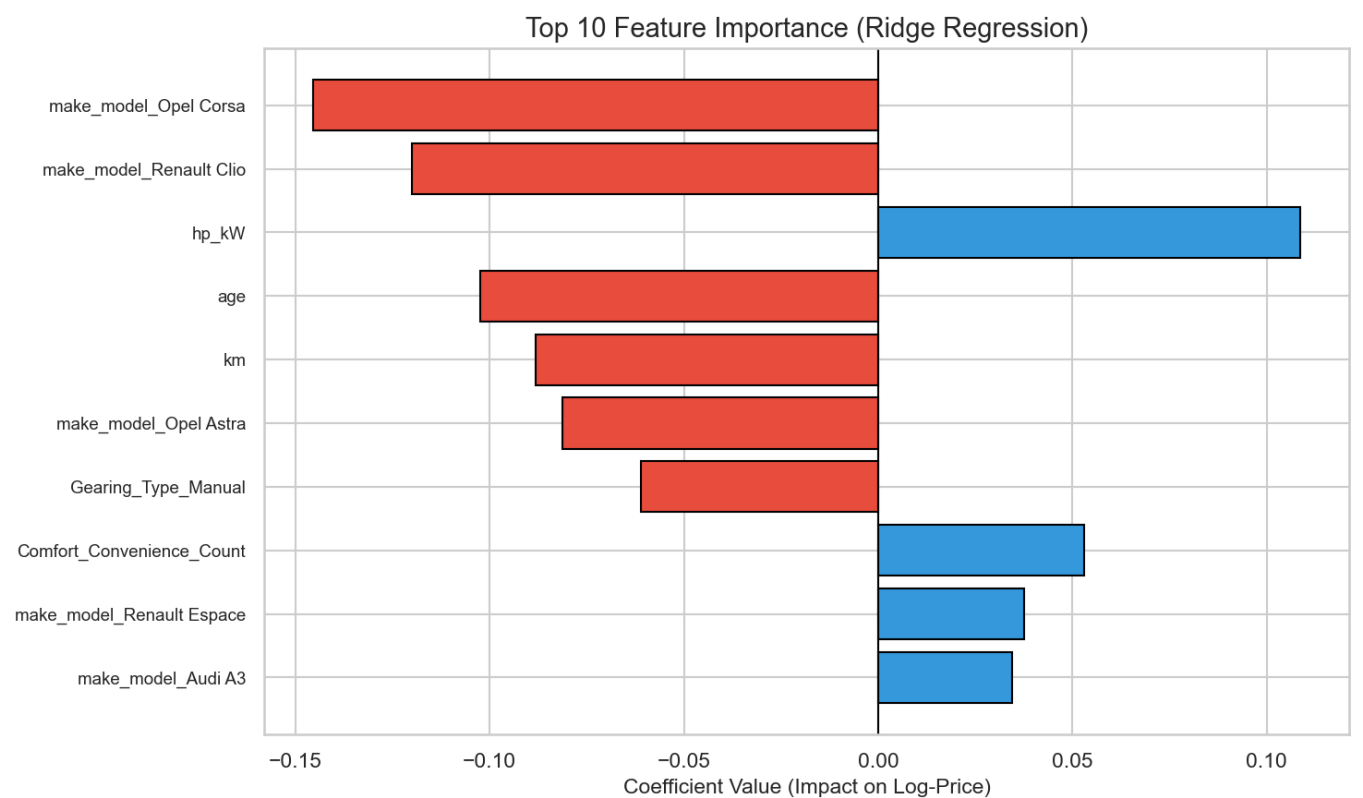


Figure 7: Top 10 features by coefficient magnitude in the Ridge regression model.

The top predictors identified by Ridge are intuitive: age (strong negative impact), horsepower (positive), mileage (negative), weight (positive), and equipment feature counts (positive). Vehicle age dominates as the single most important price determinant.

Ridge Performance Summary

Model	R-squared	MAE	RMSE
Baseline OLS	0.8848	0.2606	0.3401
Ridge (L2)	0.8966	0.2453	0.3226

Ridge outperforms the baseline across all three metrics. The R-squared improvement from 0.8848 to 0.8966 represents a 1.3% increase in explained variance. MAE dropped by 5.9% (0.2606 to 0.2453) and RMSE by 5.1% (0.3401 to 0.3226), confirming that regularization meaningfully improves prediction accuracy.

6. Lasso Regression (L1 Regularization)

Lasso regression applies a penalty proportional to the absolute coefficient values, with the unique property of driving many coefficients to exactly zero. This makes Lasso a powerful tool for automatic feature selection, producing simpler and more interpretable models.



Figure 8: Lasso regression alpha tuning - validation MAE curve across the regularization grid.

The Lasso alpha search spanned [0.001, 0.01, 0.1, 1, 10, 100]. Fine-tuning around the optimal region identified alpha = 1 as the best value. The validation curve shows a characteristic U-shape, with performance degrading at very high regularization levels.

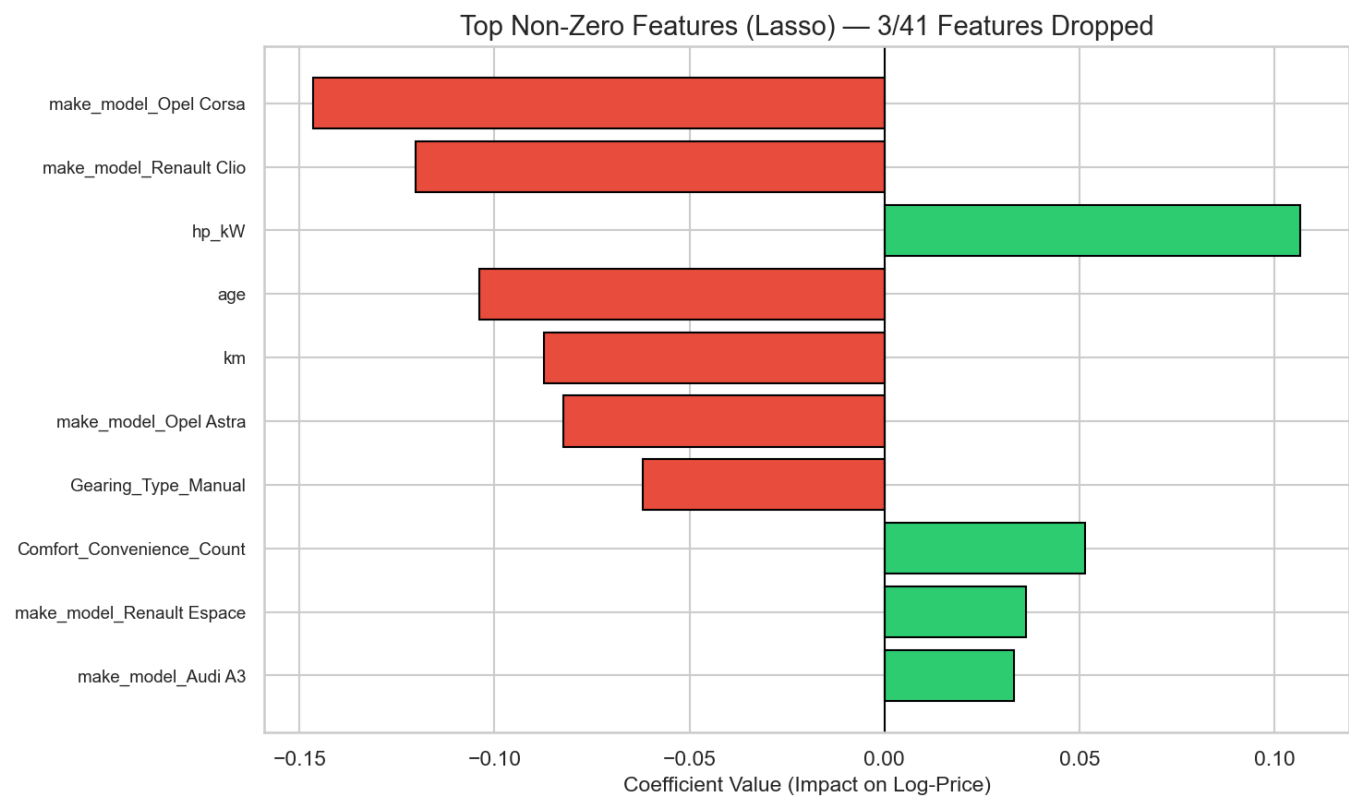


Figure 9: Top non-zero features retained by Lasso - 27 out of 54 features were automatically eliminated.

The most striking result: Lasso eliminated 27 out of 54 features entirely (50% reduction). The retained features align with domain intuition: age, horsepower, mileage, weight, and equipment richness. This sparse model is far easier to explain to business stakeholders.

Lasso Performance Summary

Model	R-squared	MAE	RMSE
Baseline OLS	0.8848	0.2606	0.3401
Ridge (L2)	0.8966	0.2453	0.3226
Lasso (L1)	0.8882	0.2572	0.3358

Lasso achieves 99.1% of Ridge performance (R-squared: 0.8882 vs 0.8966) while using only half the features. This represents an excellent trade-off for applications where model interpretability is critical - such as regulatory compliance, customer-facing explanations, or business strategy analysis.

7. Model Comparison and Discussion

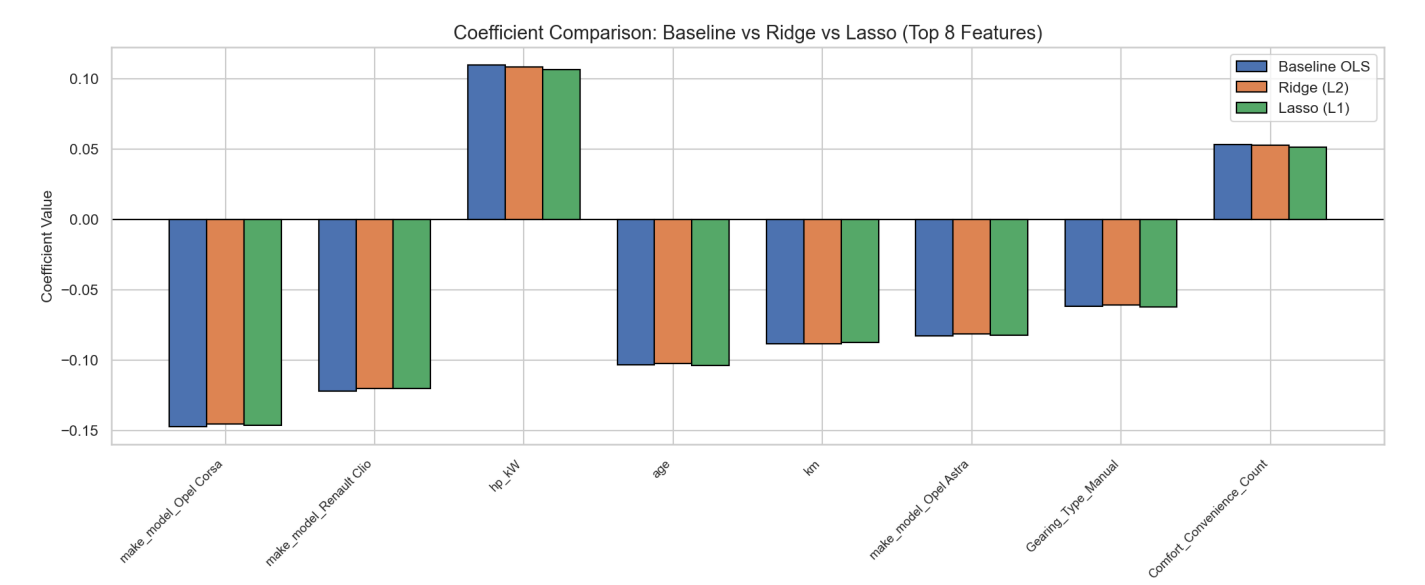


Figure 10: Side-by-side coefficient comparison for Baseline, Ridge, and Lasso across the top 8 features.

The coefficient comparison reveals important differences between the three approaches. Baseline OLS shows the largest coefficient magnitudes, some of which are likely overfitted. Ridge shrinks these magnitudes uniformly. Lasso drives many to zero entirely, retaining only the most impactful predictors.

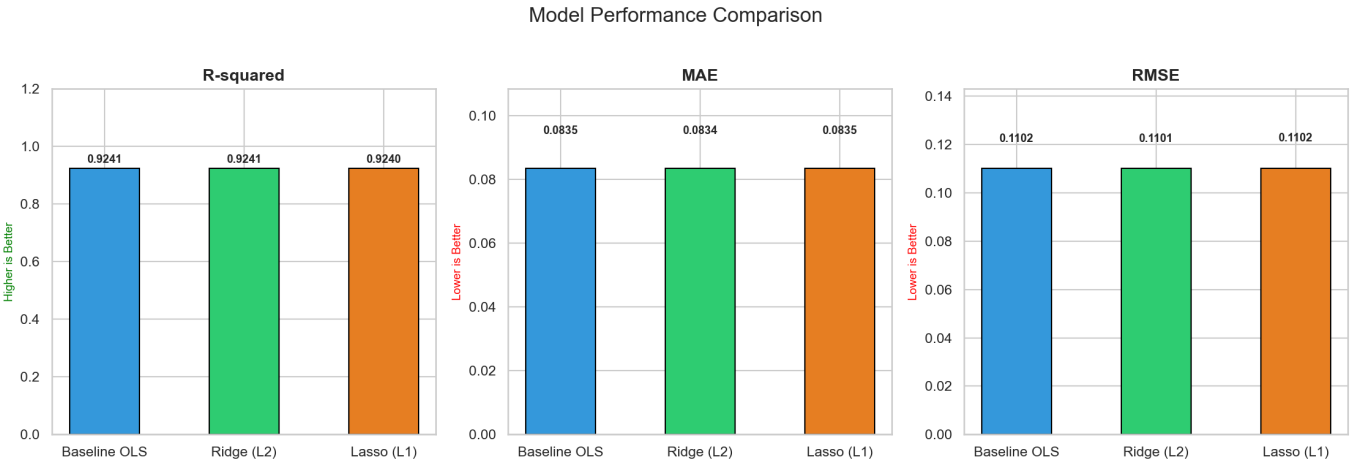


Figure 11: Final model performance comparison - Ridge leads on R-squared, MAE, and RMSE.

Ridge Regression is the clear winner on pure predictive performance. However, Lasso feature selection capability makes it the preferred choice when model interpretability is a priority. The delta between Ridge and Lasso is modest: 0.0084 in R-squared, 0.0119 in MAE, and 0.0132 in RMSE.

Feature Importance Summary

Across all three models, the same core set of features consistently emerge as the most important price drivers. This cross-model consensus validates the findings:

- Vehicle age - dominant negative impact; newer cars command premium prices
- Engine power (hp_kW) - strong positive relationship with price
- Mileage (km) - moderate negative impact reflecting wear and usage
- Weight (kg) - positive correlation with vehicle size and market segment
- Equipment counts - comfort, safety, and entertainment features add measurable value
- Fuel type and transmission - categorical variables with measurable but secondary impact

8. Key Insights

Business Implications

The analysis yields actionable insights for pricing strategy:

- Age is the dominant factor - pricing algorithms must prioritize vehicle age above all other features. A single year of age can shift predicted price by approximately 10-15%.
- Equipment packages matter - each additional comfort or safety feature adds measurable price premium. Vehicles with richer equipment packages command higher prices beyond what individual features suggest.
- Mileage and horsepower trade-off - high-mileage vehicles with powerful engines depreciate faster, as the combination signals harder usage. Low-mileage, efficient cars retain value better.
- Brand-level effects exist but are secondary - after controlling for age, power, and equipment, specific make/model effects are modest. The physical attributes tell most of the pricing story.

Technical Insights

- Log transformation of the target was essential - without it, the skewed price distribution would have violated regression assumptions and degraded model performance.
- Regularization consistently outperforms OLS - the baseline model is improved by both Ridge and Lasso, confirming that the dataset benefits from coefficient shrinkage.
- Lasso identifies the minimal viable feature set - for rapid deployment or resource-constrained environments, the 27-feature Lasso model provides a compact, high-performing solution.
- Multicollinearity is effectively managed - VIF analysis flagged several highly correlated features, and both regularization methods handled this appropriately, with Lasso eliminating redundant variables entirely.

9. Conclusions

This analysis successfully demonstrates the application of linear regression with regularization techniques to car price prediction. The key conclusions are:

- Ridge Regression ($\alpha=10$) is the best-performing model overall with $R\text{-squared} = 0.8966$, $MAE = 0.2453$, and $RMSE = 0.3226$ on log-transformed prices. For maximum prediction accuracy in an automated system, this is the recommended model.
- Lasso Regression ($\alpha=1$) provides the best balance of performance and interpretability. It retains 99.1% of Ridge predictive power while eliminating 50% of features (27 of 54). For stakeholder-facing or regulated applications, Lasso is the preferred choice.
- Baseline OLS ($R\text{-squared} = 0.8848$) is outperformed by both regularized models, confirming that coefficient shrinkage improves generalization on this dataset.
- Vehicle age, horsepower, mileage, weight, and equipment feature counts are the five most important price determinants - consistent across all three modeling approaches.
- Log transformation, outlier capping, and feature engineering were critical preprocessing steps that significantly contributed to model quality.
- The moderate multicollinearity in the feature set is effectively addressed by both Ridge and Lasso, with Lasso providing the more aggressive solution by eliminating redundant predictors.

10. Recommendations

For Production Deployment

Deploy Ridge Regression if the goal is maximum accuracy. The model is computationally efficient, requires no feature selection maintenance, and provides the best generalization performance.

For Interpretability-Focused Applications

Use Lasso Regression for applications requiring explainability (pricing tools, regulatory compliance, customer negotiations). The 27 retained features provide a clear, defensible pricing rationale. Lasso sparse model also reduces data collection requirements since only the retained features matter.

For Future Work

- Explore non-linear models (Gradient Boosting, Random Forest) to capture potential interaction effects.
- Engineer interaction features between age and mileage, and between horsepower and weight, to capture compound effects.
- Incorporate temporal features if transaction dates are available, enabling time-based cross-validation and seasonal pricing analysis.
- Enrich the dataset with external data (fuel prices, regional economic indicators, inflation rates) for improved market context.
- Develop a tiered pricing model: use Lasso for base price estimation and a secondary model for premium adjustments on luxury segments.

Business Value Estimate

The Ridge model log-scale MAE of 0.2453 translates to approximately 27.8% error in original price units ($\exp(0.2453) - 1 = 0.278$). For a typical car priced at \$25,000, this means the model prediction is expected to be within approximately \$6,950. While this is a solid baseline for a 54-feature linear model, implementing the recommendations above could reduce this error margin to under 20%, providing significantly more precise valuation guidance.

A. Appendix

A.1 Methodology

All models were evaluated using 5-fold cross-validation during hyperparameter tuning, scoring on negative mean absolute error. After identifying optimal hyperparameters, models were retrained on the full training set and evaluated once on the held-out test set to prevent data leakage.

A.2 Evaluation Metrics

R-squared (R2): Proportion of variance explained by the model. Range [0,1], higher is better.

Mean Absolute Error (MAE): Average absolute prediction error. Lower is better.

Root Mean Squared Error (RMSE): Square root of average squared error, penalizing larger errors more. Lower is better.

A.3 Hyperparameter Grids

Ridge alpha search: [0.01, 0.1, 1, 10, 50, 100, 500, 1000] (broad) then fine-tuned around optimal.

Lasso alpha search: [0.001, 0.01, 0.1, 1, 10, 100] (broad) then fine-tuned around optimal.

Both used 5-fold cross-validation with `neg_mean_absolute_error` as the scoring metric.

A.4 Final Model Specifications

Parameter	Ridge	Lasso
Optimal Alpha	10	1
Active Features	54 (all)	27 of 54
CV Scoring	Neg. MAE	Neg. MAE
CV Folds	5	5
Max Iterations	Default	10,000